

Giuseppe Abbiati

[Associate Professor in Structural Mechanics]

Dept. of Civil and Architectural Engineering (CAE), Aarhus University, Denmark

ORCID: 0000-0002-5048-8505, **Scopus ID:** 37064203300, **Scopus h-index:** 14, **Scopus citations:** 817, **Pure:** <https://pure.au.dk/portal/en/persons/abbiati@cae.au.dk>

Education

- Ph.D. in Structural Engineering, University of Trento, Italy. Doctoral thesis title: “Dynamic substructuring of complex hybrid systems based on time integration, model-order reduction and dynamic identification techniques” (April 7, 2014).
- M.Sc. in Civil Engineering, Polytechnic of Turin, Italy (December 10, 2009).
- B.Sc. in Civil and Environmental Engineering, Polytechnic of Turin, Italy (October 21, 2008).

Employments

- Associate Professor–Head of Section for Structural Engineering, Department of Engineering, Section of Civil and Architectural Engineering, University of Aarhus, Denmark (2023-now).
- Tenure-Track Assistant Professor, Department of Engineering, Section of Civil and Architectural Engineering, University of Aarhus, Denmark (2019-2023).
- Post-doctoral researcher, Chair of Structural Dynamics and Earthquake Engineering, Department of Civil, Environmental and Geomatic Engineering, ETH Zurich, Switzerland (2014-2018).

Fellowships

Recipient of Swiss Government Excellence Scholarship for a 1-year post-doctoral fellowship at the Department of Civil, Environmental and Geomatic Engineering of ETH Zurich, Switzerland, 2014, REF. 2014.0588.

Research achievements and impact

Giuseppe Abbiati’s main research contributions relate to hybrid testing and digital twins for civil engineering structures (buildings, bridges, wind parks). His cumulative portfolio of funded research and innovation projects sums up to 2.9 MEUR, of which +2 MEUR are with industrial partners; +700 kEUR currently ongoing (all related to digital twins for the built environment).

Supervision record

Current: 5 PhD students, 2 postdocs, 7 master’s students.

Total: 7 PhD students (2 already graduated), 4 postdocs, 45 master’s students.

Member of the assessment committee for 9 doctoral thesis.

Teaching experience Course responsible for 15 ECTS in the MSc (finite-element analysis, experimental mechanics, machine learning for civil engineering) and 5 ECTS in the BSc (probability and statistics) of Civil and Architectural Engineering, Aarhus University.

Selected research project

1. Enhancing Hybrid Digital Twinning for the Preservation of Built Heritage (HERITWIN), 2026-2029, Marie Curie Staff Exchange Network, European Commission 123 kEUR (1.117 MEUR total budget)
2. Laboratories as data factories for DT of floating wind turbines operational from Day 1 (LAB4TWIN) 2026-2029 CET Partnership Joint Call 2025, 334 kEUR (1.160 MEUR total budget)

3. GIS-based platform for seismic resilience of urban communities (GURU), REF. 4339-00005B 2025-2028 (36m) EUREKA – Innovation Beyond Borders - 2024-23234/NP/Resilience Call 2024 298 kEUR (1.82 MEUR total budget)
4. Cyber-Physical Sensing for Machine and Structures (CP-SENS), REF. 2081-00006B 2022-2027 (48m) Danish Innovation Fund 480 kEUR (2.55 MEUR total budget).

Selected publications

Five most important (last 15 years):

1. Scussolini, L., Abbiati, G., & Ceravolo, R. (2026). Identification of Volterra time-varying parameters in dynamical systems under random excitation. *Journal of Sound and Vibration*, 645, 120128, DOI:10.1016/j.jsv.2026.120128.
2. Pizarro, D., M. Kovarbašić, G. Abbiati, and B. Stojadinović, (2025). “Multi-Axial Sub-assembly Testing System for Hybrid Simulation with Large-Scale Structural Components.” *Earthquake Engineering & Structural Dynamics*, 54(8):2084-2105, DOI:10.1002/eqe.4351.
3. Stamenov, D., Abbiati, G., & Sauder, T. (2025). Numerical estimation of generalized frequency response functions from time series data using NARX. *Mechanical Systems and Signal Processing*, 239, 113278, DOI:10.1016/j.ymssp.2025.113278.
4. G. Abbiati, M. Broccardo, I. Abdallah, S. Marelli, F. Paolacci, (2021). Seismic Fragility Analysis based on Artificial Ground Motions and Surrogate Modeling of Validated Structural Simulators, *Earthquake Engineering and Structural Dynamics*, 50(9): 2314-2333, ISSN 00988847, DOI:10.1002/eqe.3448.
5. G. Abbiati, V. La Salandra, O.S. Bursi, L. Caracoglia (2018). A Composite Experimental Dynamic Substructuring Method Based on Partitioned Algorithms and Localized Lagrange Multipliers. *Mechanical Systems and Signal Processing*, 100:85-112, ISSN 08883270, DOI:10.1016/j.ymssp.2017.07.020.
6. Abbiati G., R. Ceravolo, C. Surace (2015). Time-dependent Estimators for On-Line Monitoring of Full-Scale Structures under Ambient Excitation, *Mechanical Systems and Signal Processing*, 60:166-181, ISSN 08883270, DOI:10.1016/j.ymssp.2014.10.018.

Five recent, project-relevant (last 5 years):

1. P. Talasila, D. Tcherniak, A. M.D. Jensen, S Mahato, J. V. Medom, M. D. Ulriksen, G. Abbiati, A. Schörghofer-Queiroz, P. G. Larsen, L. Damkilde, (2026). A Digital Twin Platform for Structural Health Monitoring, *Computer-Aided Civil and Infrastructure Engineering*, 50:100086, ISSN 1093-9687, DOI:10.1016/j.cacaie.2026.100086.
2. Hansen, S. T., Thule, C., Gomes, C., Lausdahl, K. G., Madsen, F. P., Abbiati, G., & Larsen, P. G. (2024). Co-simulation at different levels of expertise with Maestro2. *Journal of Systems and Software*, 209, 111905, DOI:10.1016/j.jss.2023.111905.
3. Oakes, B, C Gomes, G Abbiati, E Kamburjan, E E. Baş, and S Engelsgaard. “Towards Ontological Service-Driven Engineering of Digital Twins.” In 1st International Conference on Engineering Digital Twins (EDTconf 2024). Linz, Austria: to appear, 2024, DOI:10.1145/3652620.3688261.
4. Baş, E. E., Abbiati, G., Gonçalves Gomes, C. Â., Jassmann, U., & Larsen, P. G. (2024). Digitalization of Large-Scale Testing Facilities for the Wind Industry: DIGIT-BENCH Digital Twin. *Journal of Physics: Conference Series*, 2767(4), 042033, DOI:10.1088/1742-6596/2767/4/042033.
5. Abbiati, G., C. Gomes, M. Sandberg, Z. Kazemi, S. T. Hansen, and P. G. Larsen. “Modelling for Digital Twins.” In *The Engineering of Digital Twins*, edited by J. Fitzgerald, C. Gomes, and P. G. Larsen, 89–127. Cham: Springer International Publishing, 2024. DOI:10.1007/978-3-031-66719-0_5.